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Experiment 1

Collect a real-life dataset and identify different types of data(numerical and categorical). Analyse the dataset and present it using suitable tables or charts.

1.1. Dataset Info

```
import pandas as pd
import matplotlib.pyplot as plt
import random

df_student = pd.read_csv("dataset.csv", delimiter=',',
    header='infer')

df_student.head()
```

Output:

	Name	Gender	Grade	StudyHours	Marks
0	Aarav	M	A	6	91
1	Ishita	F	A	7	95
2	Rohan	M	B	4	75
3	Meera	F	B	5	78
4	Kabir	M	C	2	55

`head()` function is used to print the first 5 records of the dataframe. We can use `dtypes` attribute to give out datatype of each column.

```
df_student.dtypes
```

Output:

```
Name           object
Gender         object
Grade          object
StudyHours      int64
Marks           int64
Attendance      int64
dtype: object
```

The dataset we chose contains 50 records. These records were Name, Gender, Grade, StudyHours, Marks & Attendance.

- **Numerical:** StudyHours, Marks, Attendance
- **Categorical:** Gender (M/F), Grade (A/B/C/D)

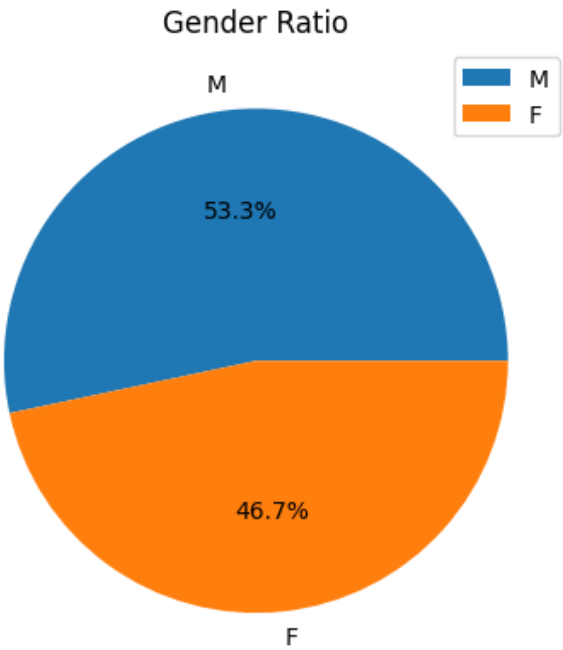
1.2. Analysis of Dataset

Let's do some basic analysis of the dataset using charts and graphs.

i) Gender Distribution

```
counts = df_student.Gender.value_counts()
plt.pie(counts, labels=counts.index, autopct='%1.1f%%')
plt.legend()
plt.title("Gender Ratio")
plt.show()
```

Output:

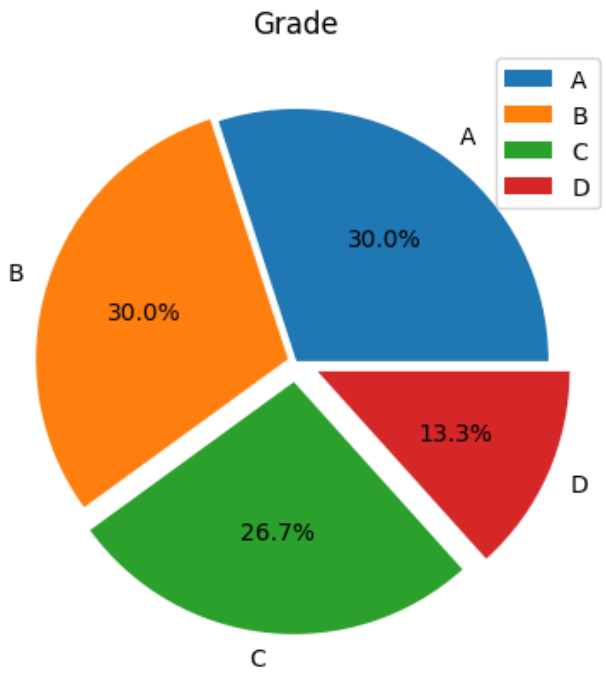


ii) Grade Distribution

```
counts = df_student.Grade.value_counts()
plt.pie(counts, labels=counts.index, autopct='%1.1f%%',
    explode=[random.uniform(0, 0.1) for _ in range(4)])
plt.legend()

plt.title("Grade")
plt.show()
```

Output:

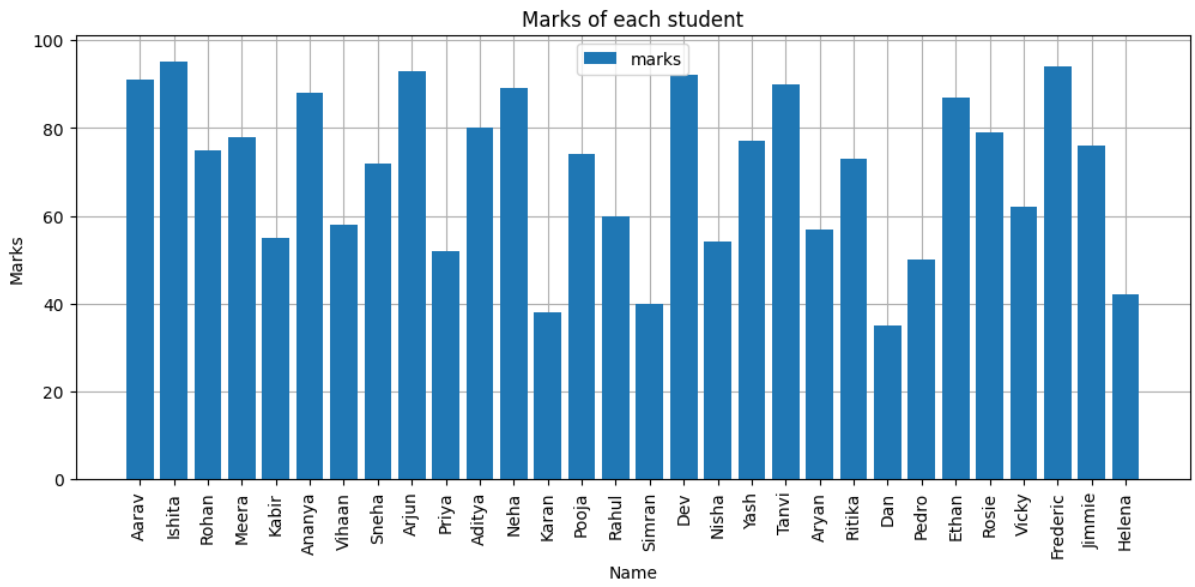


iii) Marks of Each Student

```
plt.figure(figsize=(12,4.8))
plt.bar(df_student.Name, df_student.Marks, label="marks",
    zorder=2)

plt.title("Marks of each student")
plt.ylabel("Marks")
plt.xlabel("Name")
plt.ylim(top=101)
plt.xticks(rotation=90)
plt.grid(zorder=0)
plt.legend()

plt.show()
```



iv) Gender-wise average distribution

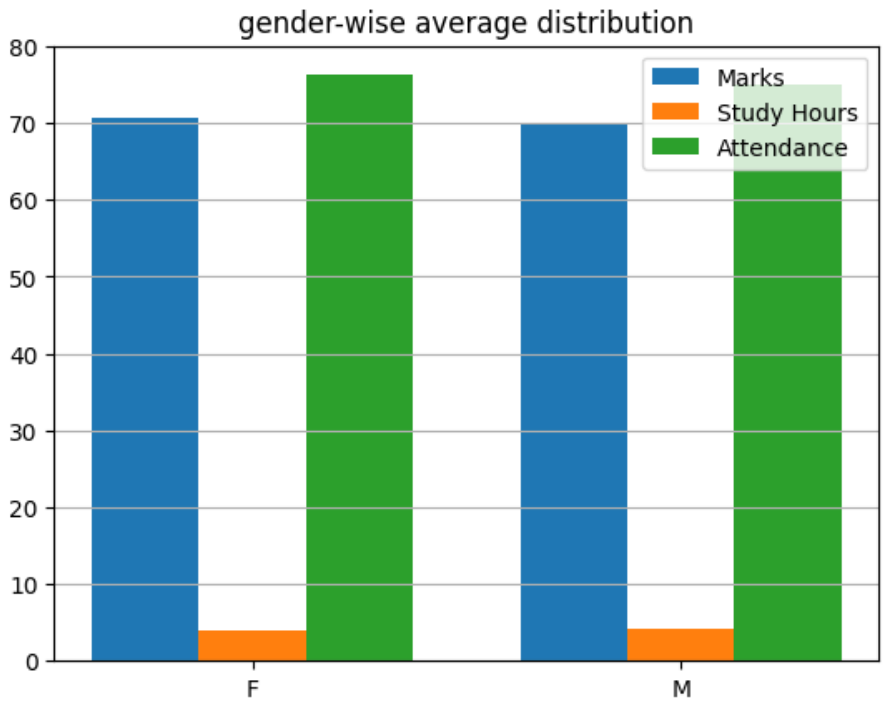
```
marks = df_student.groupby("Gender")["Marks"].mean()
studyhour = df_student.groupby("Gender")["StudyHours"].mean()
hardiksassignment =
    df_student.groupby("Gender")['Attendance'].mean()

x = np.arange(marks.count())
width = 0.25

plt.bar(x-width, marks.values, width, label="Marks")
plt.bar(x, studyhour.values, width, label="Study Hours")
plt.bar(x + width, hardiksassignment.values, width,
    label="Attendance")

plt.xticks(x, marks.index)
plt.grid(axis='y', zorder=0)
plt.legend()
plt.title("gender-wise average distribution")

plt.show()
```



v) Grade-wise average distribution

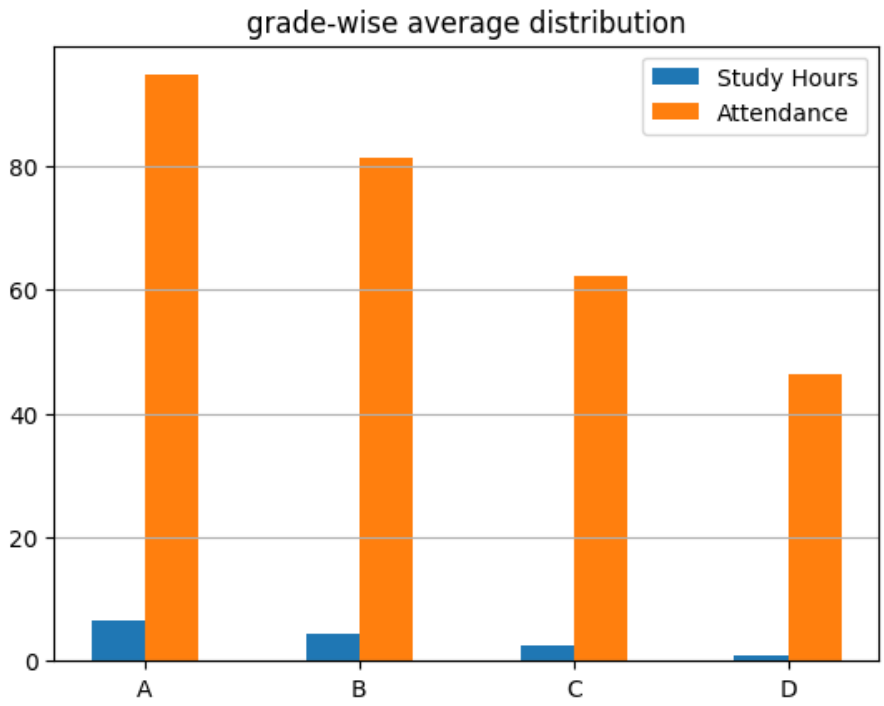
```
studyhour = df_student.groupby("Grade")["StudyHours"].mean()
hardiksassignment =
    df_student.groupby("Grade")['Attendance'].mean()

x = np.arange(marks.count())
width = 0.25

plt.bar(x - width/2, studyhour.values, width, label="Study
    Hours")
plt.bar(x + width/2, hardiksassignment.values, width,
    label="Attendance")

plt.xticks(x, marks.index)
plt.grid(axis='y', zorder=0)
plt.legend()
plt.title("grade-wise average distribution")

plt.show()
```



1.3. Observation

From these graphs and charts, we can deduce that this batch has an approximately equal ratio of males and females. Both groups have fairly similar average marks, study hour and attedance. Looking at the grade-wise distribution, we can see that as the grade descreases the study hours and attendance have also reduced.

Experiment 2

Take a dataset and classify the data based on measurement scales(nominal, ordinal, internval, and ratio). Provide appropriate examples for each category.

2.1. Dataset Info

```
import pandas as pd
import matplotlib.pyplot as plt
import random

df_student = pd.read_csv("dataset.csv", delimiter=',',
    header='infer')

df_student.head()
```

Output:

	Name	Gender	Grade	StudyHours	Marks
0	Aarav	M	A	6	91
1	Ishita	F	A	7	95
2	Rohan	M	B	4	75
3	Meera	F	B	5	78
4	Kabir	M	C	2	55

`head()` function is used to print the first 5 records of the dataframe. We can use `dtypes` attribute to give out datatype of each column.

```
df_student.dtypes
```

Output:

```
Name           object
Gender         object
Grade          object
StudyHours      int64
Marks           int64
Attendance      int64
dtype: object
```

The dataset we chose contains 50 records. These records were Name, Gender, Grade, StudyHours, Marks & Attendance.

- **Numerical:** StudyHours, Marks, Attendance
- **Categorical:** Gender (M/F), Grade (A/B/C/D)

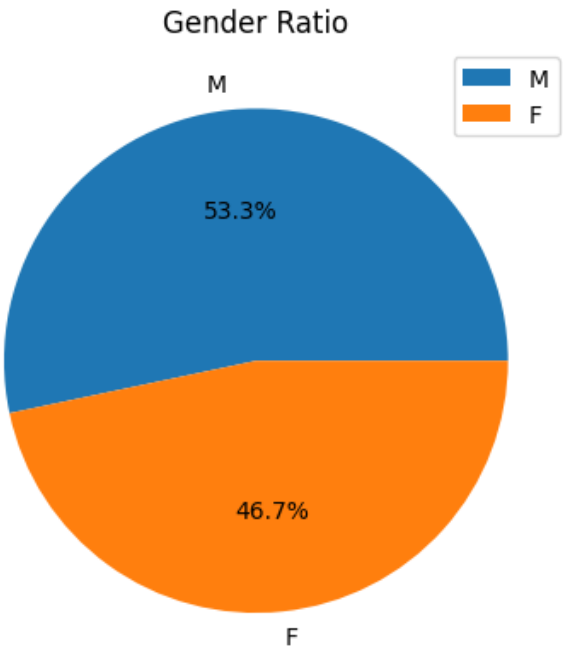
2.2. Analysis of Dataset

Let's do some basic analysis of the dataset using charts and graphs.

i) Gender Distribution

```
counts = df_student.Gender.value_counts()
plt.pie(counts, labels=counts.index, autopct='%1.1f%%')
plt.legend()
plt.title("Gender Ratio")
plt.show()
```

Output:

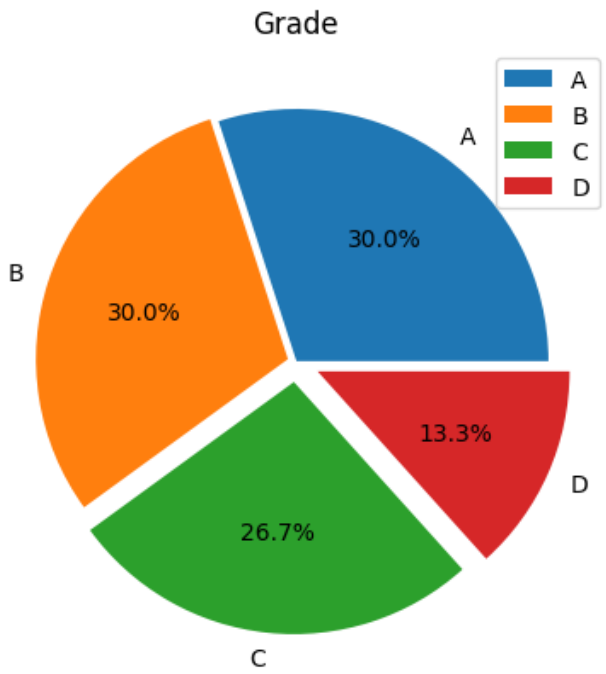


ii) Grade Distribution

```
counts = df_student.Grade.value_counts()
plt.pie(counts, labels=counts.index, autopct='%1.1f%%',
    explode=[random.uniform(0, 0.1) for _ in range(4)])
plt.legend()

plt.title("Grade")
plt.show()
```

Output:

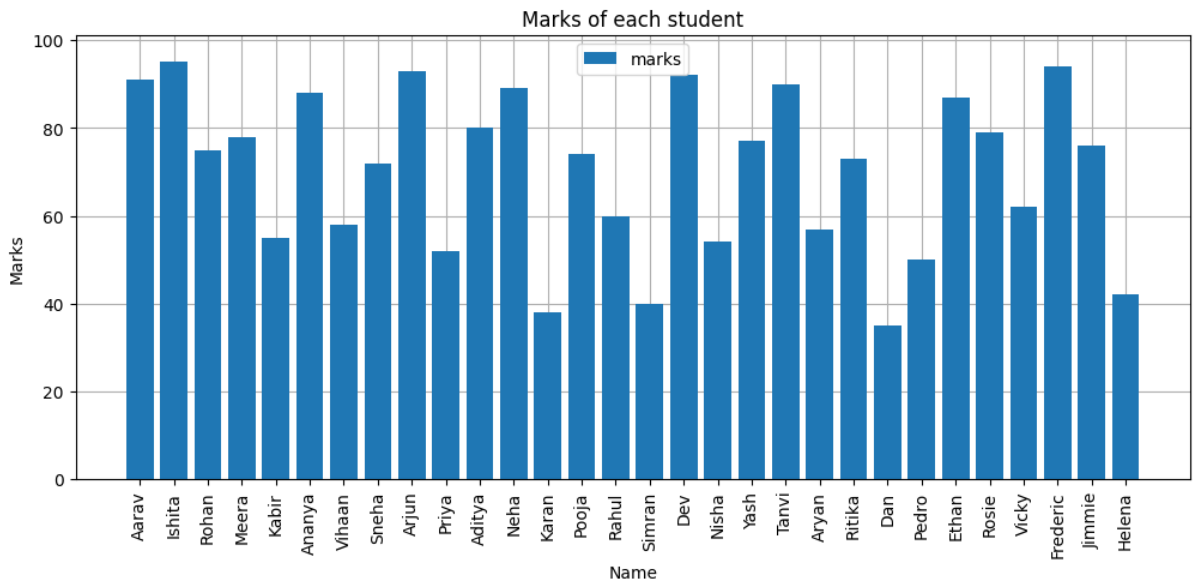


iii) Marks of Each Student

```
plt.figure(figsize=(12,4.8))
plt.bar(df_student.Name, df_student.Marks, label="marks",
    zorder=2)

plt.title("Marks of each student")
plt.ylabel("Marks")
plt.xlabel("Name")
plt.ylim(top=101)
plt.xticks(rotation=90)
plt.grid(zorder=0)
plt.legend()

plt.show()
```



iv) Gender-wise average distribution

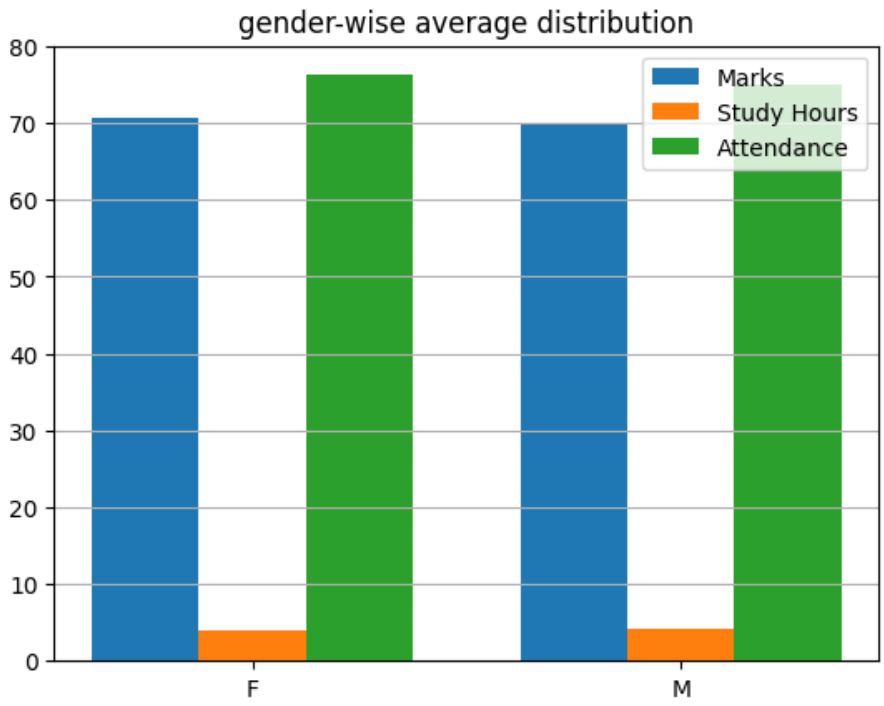
```
marks = df_student.groupby("Gender")["Marks"].mean()
studyhour = df_student.groupby("Gender")["StudyHours"].mean()
hardiksassignment =
    df_student.groupby("Gender")['Attendance'].mean()

x = np.arange(marks.count())
width = 0.25

plt.bar(x-width, marks.values, width, label="Marks")
plt.bar(x, studyhour.values, width, label="Study Hours")
plt.bar(x + width, hardiksassignment.values, width,
    label="Attendance")

plt.xticks(x, marks.index)
plt.grid(axis='y', zorder=0)
plt.legend()
plt.title("gender-wise average distribution")

plt.show()
```



v) Grade-wise average distribution

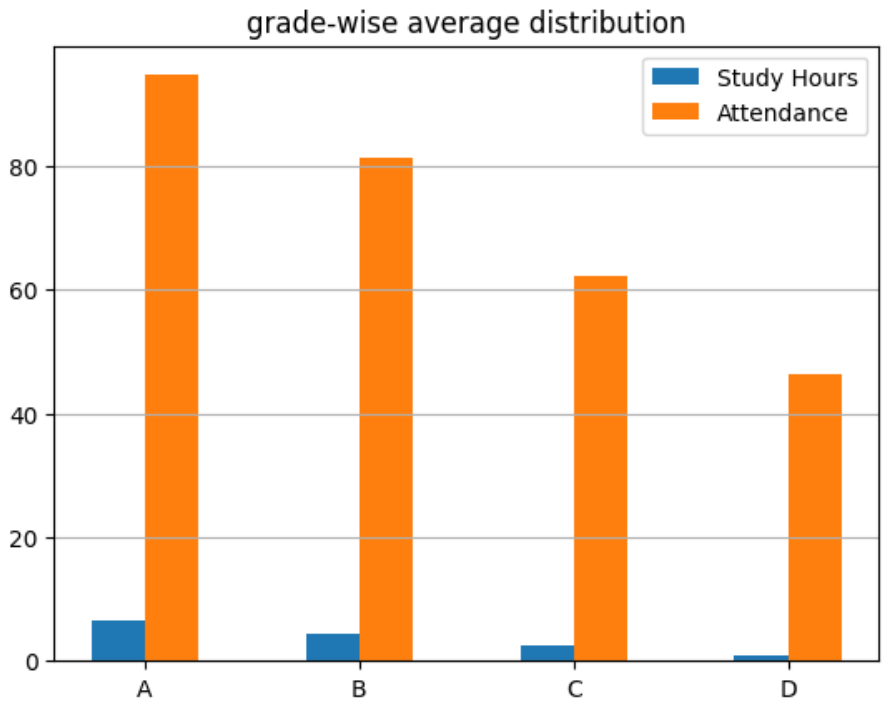
```
studyhour = df_student.groupby("Grade")["StudyHours"].mean()
hardiksassignment =
    df_student.groupby("Grade")['Attendance'].mean()

x = np.arange(marks.count())
width = 0.25

plt.bar(x - width/2, studyhour.values, width, label="Study
    Hours")
plt.bar(x + width/2, hardiksassignment.values, width,
    label="Attendance")

plt.xticks(x, marks.index)
plt.grid(axis='y', zorder=0)
plt.legend()
plt.title("grade-wise average distribution")

plt.show()
```



2.3. Observation

From these graphs and charts, we can deduce that this batch has an approximately equal ratio of males and females. Both groups have fairly similar average marks, study hour and attedance. Looking at the grade-wise distribution, we can see that as the grade descreases the study hours and attendance have also reduced.